# TANBIR RAHMAN

Eligible for Engineer-in-Training (EIT)

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## SUMMARY


Master of Engineering in Electrical & Computer Engineering at the University of Calgary (**graduating Apr 2026**) with specialization in power systems. Experienced in transmission system studies including power flow, contingency analysis, and grid congestion using PSS®E. Skilled in stability analysis, renewable integration, and automated simulation workflows using Python. Seeking a transmission planning and interconnection studies role.


## TECHNICAL SKILLS

- Power System Studies: Load flow analysis, Short-circuit studies, Arc-flash analysis, Voltage Stability, Congestion Analysis, Transient stability studies, protection co-ordination.
- Documentation: Microsoft Office (Excel, Word, PowerPoint), Bluebeam, Power BI.
- Engineering Tools: PSSE, PSCAD, Python, AutoCAD, ETAP, MATLAB/Simulink, ArcGIS.
- Standards: Canadian Electrical Code (CEC), IEEE std 80, IEEE 1547-2018, TPL Standards (NERC and AESO), NFPA 70E.

## EDUCATION

**Master of Engineering, Electrical and Computer Engineering** 2024-Apr 2026
Institute: University of Calgary

**Bachelor of Science, Electrical and Electronic Engineering** 2018-2022
Institute: Shahjalal University of Science & Technology

## SELECTED PROJECTS

**Automated Grid Congestion, voltage violation & Curtailment Screening Tool for Alberta Transmission Network**

*Tools: PSSE, Python, Excel*

- Developed automated framework to run 8,760 hourly AC power flow simulations using PSS®E
- Performed transmission system analysis including contingency screening (N-0, N-1) and voltage violation assessment
- Evaluated grid congestion and generator curtailment under varying system and renewable generation conditions
- Automated reporting using Python, reducing manual analysis time by over 90%
- Generated technical reports aligned with transmission planning and reliability requirements

**DER Interconnection Study & Compliance Analysis (FortisAlberta Standards)**

*Tools: PSSE, Excel*

- Conducted DER interconnection study for generator connection to utility network using PSS®E
- Performed short-circuit and effective grounding studies to ensure compliance with utility requirements
- Evaluated protection system performance, including relay desensitization and coordination under fault conditions
- Assessed interconnection impacts on system reliability and developed updated SLD with mitigation solutions
- Applied FortisAlberta DER standards (DER-02 series) and IEEE 1547; prepared technical report with protection settings and SCADA/DNP point list

**Transmission Line Fault Detection and Classification using ML**

Tools: MATLAB, Excel, Latex, Origin, Python

- Developed a Simulink transmission line model to simulate multiple fault scenarios.
- Extracted voltage/current features and trained ML classifiers using Python.
- Achieved ~99% accuracy in identifying transmission line fault types.

## EXPERIENCE

**Project Assistant - University Research Centre, SUST** 2022-2023

- Assisted in development of an IoT-based solar-powered smart vehicle system,
- Supported power conversion design and optimization, improving system efficiency to 82%.
- Collaborated with research team to evaluate system performance and reliability.

## Professional Development

- **Load Flow and Stability Analysis** – Electro Mentors
- **EMT of Transmission Lines** – Electro Mentors
- **HVDC Systems with EMT Software Tool** – Electro Mentors
- **Fundamentals of EMT Studies, IBRs and Large Loads** – Electro Mentors